Individualism and Innovation:

Unraveling Culture's Influence Across Time*

Rouzhi Liang†

Latest version available here

Abstract

This paper examines the impact of individualism on innovation and economic development over time. I begin by extending the cross-country literature, using the same time-invariant cultural measures but incorporating a longer time horizon and additional control variables. The estimated effects of individualism are inconsistent across specifications and time periods, particularly when I control for institutions. To address the limitations of time-invariant measures, I then turn to a within-U.S. analysis and construct time-varying, ancestry-based measures of individualism. This variation enables the use of fixed effects to capture local institutions and isolate the influence of culture. The results suggest that individualism either has a negative impact or no significant effect on innovation and economic development, in contrast to the prevailing findings in the existing literature.

Keywords: culture, innovation, economic development, economic growth

JEL Classification: O47 Z13

*I sincerely thank my advisor, Dr. Ömer Özak, for his invaluable guidance and support throughout this research.

†Department of Economics, Southern Methodist University. E-mail: rouzhil@smu.edu

1 Introduction

Existing cross-country evidence suggests that individualism promotes innovation and economic development by increasing incentives for technological innovation (Gorodnichenko and Roland, 2011a,b). However, this conclusion requires further discussion, given two important concerns. First, it only focuses on a short time period with a fixed measure of individualism, ignoring the fact that the influence of individualism evolves over time as societies develop. Second, because the measures of individualism are fixed at the country level, it's hard to disentangle the effect of culture from that of formal institutions such as the patent law. In the empirical analysis, this prevents the use of country fixed effects. As a result, studies often use institutional indices to account for institutions, which may interact with cultural traits and generate biased estimation.

This paper reassesses the influence of individualism on innovation and economic development by addressing those concerns. I first extend the previous cross-country studies by using a longer time scope and a richer set of controls. To better isolate the effect of individualism from that of institutions, I also conduct a within-US analysis and construct a time-varying proxy for individualism based on ancestry composition. This within-country approach allows for variation in culture at the county-group level with a common national institutional background, enabling the inclusion of local fixed effects and helping to better distinguish cultural influences from that of institutions.

The results challenge the conventional view. In the cross-country regressions, the sign, significance, and magnitude of individualism's influence are highly sensitive to the specification, especially when institutional factors are considered. Moreover, the influence of individualism changes across different time periods. In the within-US analysis, where national institutions are fixed, and local institutions can be controlled using local fixed effects, individualism either has no significant effect on, or negatively affects, innovation and economic development.

Understanding why the findings of this paper differ from the existing literature is important. This paper's results suggest that individualism's influence may depend on historical and institutional contexts rather than being constant and universal. In the cross-country analysis, individualism can be correlated with institutional features such as property rights and legal origins, which can influence economic performance by themselves. Moreover, if individualism level and the impact of individualism change over time, estimation based on time-invariant cultural indexes within a short time period may obscure important dynamics. From this perspective, the positive influence of individualism documented in the previous literature may partially reflect institutional effects and may not be robust over time.

This paper contributes to several strands of the literature. First, it contributes to the literature on culture and economic development by reassessing the role of individualism using new empirical strategies that address both time variation and institutional confounding. Second, it relates to the literature on subnational cultural variation and its impact, showing how within-country analysis can provide insights that are difficult to obtain from cross-country comparisons. Third, it contributes to the literature on immigration and local economic development by using ancestry composition to construct time-varying measures of individualism and study its economic influence.

The rest of the paper is structured as follows. In Section 2, I review related literature. In Section 3, I revisit the analysis exploring the impact of individualism on economic development, TFP, and innovation across countries with an extended time scope and more controls. In Section 4, I explain the analysis within the US. Section 5 concludes.

2 Related Literature

A large body of literature studies the relationship between culture and economic development and emphasizes how cultural values shape individual behavior and, in turn, influence economic performance through factors such as trust, religions, long-term orientation, and family ties (Alesina and Giuliano, 2014; Andersen et al., 2017; Becker and Woessmann, 2009; Dohmen et al., 2015; Galor and Özak, 2016; Zak and Knack, 2001). The most closely related papers are Gorodnichenko and Roland (2011a,b), which document a positive influence of individualism on innovation and economic development in cross-country data.

Despite these findings, the cross-country literature faces important limitations. Cultural measures are typically treated as time-invariant, which prevents the inclusion of country fixed effects and makes it difficult to separate the effects of culture from those of institutions and other persistent country characteristics. In addition, the use of time-invariant measures does not allow researchers to capture changes in cultural values or to examine how the influence of culture evolves over time. As a result, the estimated relationship between individualism and economic outcomes may be confounded by unobserved country characteristics, while being unable to capture how cultural values and their effects evolve over time.

Existing studies have attempted to address these concerns using institutional controls and instrumental variables. In cross-country settings, institutional indices are often included to account for differences in formal institutions. While useful, these controls are unlikely to capture the full set of persistent country-level factors that may be correlated with both culture and economic outcomes. Studies have also used alternative measures and instrumental variables for individualism. In particular, linguistic features such as pronoun-drop structure and the prevalence of historical disease have been used as proxies or instruments for individualism (Davis and Abdurazokzoda, 2016; Kashima and Kashima, 1998; Murray and Schaller, 2010). These approaches are useful because they move beyond contemporaneous survey-based measures and help alleviate some concerns related to reverse causality and measurement error. At the same time, they may still remain correlated with other determinants of economic development, such as institutions or deeper historical conditions. As a result, they do not fully resolve the difficulty of isolating the effect of culture in cross-country settings.

Building on this literature, this paper reassesses the cross-country relationship between individualism and economic outcomes using a longer time horizon and a richer set of controls. The relationship becomes sensitive to specifications, highlighting the difficulty of isolating the effect of individualism from other correlated factors in a cross-country setting.

These challenges motivate a complementary within-country approach. Some studies exploit within-country variation, where institutional differences can be more effectively controlled using fixed effects, and identification can rely on localized variation (Fulford et al., 2020; Sequeira et al., 2020). Within-country analyses, therefore, provide a useful setting for reassessing the relationship between culture and economic outcomes, especially under a common national institutional environment.

In this paper, I exploit variation in individualism across U.S. county groups. Following Fulford et al. (2020), I construct time-varying measures of individualism based on ancestry composition, building on the idea that cultural traits are transmitted across generations (Giavazzi et al., 2019). This approach introduces variation in culture across places and time, allowing for the inclusion of local fixed effects and enabling the analysis of time-varying cultural influences.

Taken together, the paper contributes by first reassessing the influence of individualism across countries using a more comprehensive empirical framework, and then providing complementary evidence with time-varying measures of individualism within the US.

3 Cross-country Analysis

This section reassesses the influence of individualism on innovation and economic development across countries. The main goal of this section is to examine whether the influence of individualism remains robust to a longer period, additional controls, instrument variables, and different proxies. I show that the cross-country evidence is highly sensitive to specification and unstable across time, which motivates the within-U.S. analysis in the next section.

3.1 Data and Measurement

I begin with Hofstede's index of individualism at the country-level, the standard measure used in the existing cross-country literature (Hofstede et al., 2005). Because this measure is contemporary and time-invariant, it raises two concerns. First, reverse causality may arise if economic development affects cultural values. Second, time-invariant measures make it difficult to distinguish the effect of culture from that of institutions in panel settings.

To address concerns about reverse causality, I also consider historical proxies and instruments for individualism based on pronoun-drop language structure (Davis and Abdurazokzoda, 2016; Kashima and Kashima, 1998) and historical disease prevalence (Murray and Schaller, 2010). These measures are predetermined before modern economic outcomes and are therefore less likely to be influenced by modern economic development.

At the same time, these historical variables may still be correlated with institutional characteristics, since both culture and institutions are shaped by historical factors. As a result, the IV and proxy specifications should be interpreted primarily as robustness checks rather than as providing clean identification of the causal effect of culture.

The dependent variables are log GDP per capita, total factor productivity (TFP), and log patents per capita. Following Gorodnichenko and Roland (2017), I use GDP per capita as a measure of economic development, TFP as a measure of productivity, and patents as a proxy for innovation. The control variables include geography, religion, legal origins, and property rights.

3.2 Empirical Strategy

3.2.1 Baseline Specifications

I begin with baseline OLS specifications to document the cross-country correlation between individualism and economic outcomes.

$$Y_{it} = \alpha_0 + \alpha_1 Idv_i + \boldsymbol{\alpha_2 X_{it}} + \gamma_t + \phi_{continent} + \omega_{t,continent} + \epsilon_{it}, \tag{1}$$

where Y_{it} denotes log GDP per capita, TFP, or log patents per capita for country i in year t, Idv_i is Hofstede's individualism index, and $\boldsymbol{X}_{it}$ includes the control variables. The specification includes year fixed effects, continent fixed effects, and continent-by-year interactions.

These estimates from OLS may be biased due to reverse causality and omitted variables, such as institutional differences. To address endogeneity concerns, I next employ instrumental variables based on linguistic structure and historical disease prevalence. These instruments aim to isolate exogenous variation in individualism that is less likely to be influenced by contemporary economic conditions.

While the instrumental variables strategy based on historical measures helps mitigate concerns about reverse causality and omitted variable bias, it does not fully resolve the identification problem. In particular, the validity of the instruments relies on the exclusion restriction, which may be violated if linguistic structure or historical disease environments also influence economic development through channels other than culture, such as institutions.

Therefore, the IV results should be interpreted with caution. Rather than providing causal estimates, they serve as exercises that assess the robustness of the relationship between individualism and economic outcomes across different specifications and identification strategies.

Finally, I consider historical proxies for individualism. Unlike the modern survey-based measure of individualism, these proxies are predetermined and reflect deeper, long-standing cultural traits. As such, they help mitigate concerns related to reverse causality and measurement error arising from individual subjective factors, and allow me to assess whether the results are robust across alternative measures of culture. However, as with the IV approach, these proxies may still be correlated with other country characteristics, such as institutions, and therefore do not fully resolve identification concerns.

3.3 Results

3.3.1 Economic Development

The positive cross-country influence of individualism on GDP per capita is not robust across specifications. In baseline OLS regressions, individualism has a positive effect on GDP. However, the magnitude declines substantially once additional controls, particularly institutional variables, are included.

When instruments are used, the impact of individualism starts to lose statistical significance. However, the coefficient increases when geographic factors, religions, and property rights are taken into account. On the other hand, when the pronoun drop is used as a proxy, individualism's influence is not significant once all controls are included. In other cases, the magnitude of individualism's influence drops. However, when religions are controlled, the impact of individualism increases. These patterns suggest that the relationship between individualism and economic development is unstable, especially when institutional factors are taken into account.

Table 1: Individualism and Economic Development across countries

	Log GDP per capita (1789-2018)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: OLS Regression						
Individualism	0.412***	0.336***	0.297***	0.415***	0.345***	0.194**
	(0.08)	(0.09)	(0.07)	(0.08)	(0.08)	(0.09)
Adjusted-R^2	0.775	0.783	0.818	0.776	0.789	0.836
Observations	6584	6584	6584	6584	6584	6584
Panel B: IV = DA						
Individualism	0.952***	1.059*	0.722*	0.980***	0.970**	0.915
	(0.32)	(0.54)	(0.40)	(0.34)	(0.43)	(1.11)
1st Stage F	5.20	3.42	1.90	5.37	3.43	0.91
Observations	6584	6584	6584	6584	6584	6584
Panel C: Proxy = DA						
Individualism	0.544***	0.400**	0.302*	0.559***	0.438***	0.157
	(0.15)	(0.15)	(0.16)	(0.15)	(0.14)	(0.13)
Adjusted-R^2	0.747	0.798	0.796	0.751	0.770	0.848
Observations	6584	6584	6584	6584	6584	6584
Continent FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Continent × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Geo controls	No	Yes	No	No	No	Yes
Legal Origins	No	No	Yes	No	No	Yes
Property Rights	No	No	No	No	Yes	Yes
Religions	No	No	No	Yes	No	Yes

Note: Panels A and B use the standardized individualism measure as the main explanatory variable. Panel B reports 2SLS estimates using DA as an instrument for individualism. Panel C replaces the individualism measure with DA directly as a proxy. All specifications include continent, year, and continent × year fixed effects. Column (2) adds geographic controls; column (3) uses legal-origin controls; column (4) uses religion controls; column (5) uses property-rights controls; and column (6) includes all controls. Standard errors are two-way clustered at the country and year levels. The reported first-stage statistic is the weak-identification F-statistic. *$p < 0.10$, ** $p < 0.05$, ***$p < 0.01$.

3.3.2 Total Factor Productivity

The results for total factor productivity show a similar lack of consistency and robustness. While individualism is positively associated with TFP in baseline OLS regressions, the effect weakens or disappears once additional controls are introduced. The IV and proxy estimates are generally insignificant. Overall, the evidence does not support a stable relationship between individualism and productivity across countries.

3.3.3 Innovation

The results for innovation are likewise unstable. Individualism is positively correlated with patenting activity in baseline OLS regressions, but the magnitude declines once controls are included. The IV and proxy estimates are not robust across specifications, and the impact of individualism only increases and stays significant when geographic characteristics are included in the IV regression. These findings suggest that the cross-country evidence does not show a stable relationship between individualism and innovation.

3.4 Additional Evidence

To further assess the robustness of the relationship, I present additional descriptive evidence in this section.

First, Figure 1 shows binned scatter plots with and without controls. The positive association between individualism and economic development becomes substantially flatter once controls are included, showing the sensitivity of the results to different specifications.

Second, I examine whether the relationship varies over time by including fixed effects for the interaction between individualism and year. Figure 2 shows that the association

Table 2: Individualism and TFP across countries

	Total Factor Productivity (1954-2019)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: OLS Regression						
Individualism	0.069**	0.009	0.055	0.063*	0.063*	-0.005
	(0.03)	(0.04)	(0.04)	(0.03)	(0.03)	(0.04)
Adjusted-R^2	0.17	0.28	0.26	0.28	0.19	0.37
Observations	3602	3602	3602	3602	3602	3602
Panel B: IV = DA						
Individualism	0.133	0.075	0.186	0.153	0.117	0.186
	(0.11)	(0.15)	(0.15)	(0.11)	(0.11)	(0.27)
1st Stage F	6.73	6.13	2.89	6.55	6.20	2.65
Observations	3602	3602	3602	3602	3602	3602
Panel C: Proxy = DA						
Individualism	0.081	0.038	0.092	0.095	0.069	0.056
	(0.06)	(0.07)	(0.08)	(0.07)	(0.06)	(0.07)
Adjusted-R^2	0.15	0.37	0.26	0.28	0.17	0.42
Observations	3602	3602	3602	3602	3602	3602
Continent FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Continent × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Geo controls	No	Yes	No	No	No	Yes
Legal Origins	No	No	Yes	No	No	Yes
Property Rights	No	No	No	No	Yes	Yes
Religions	No	No	No	Yes	No	Yes

Note: Panels A and B use the standardized individualism measure as the main explanatory variable. Panel B reports 2SLS estimates using DA as an instrument for individualism. Panel C replaces the individualism measure with DA directly as a proxy. All specifications include continent, year, and continent × year fixed effects. Column (2) adds geographic controls; column (3) uses legal-origin controls; column (4) uses religion controls; column (5) uses property-rights controls; and column (6) includes all controls. Standard errors are two-way clustered at the country and year levels. The reported first-stage statistic is the weak-identification F-statistic. *$p < 0.10$, ** $p < 0.05$, ***$p < 0.01$.

between individualism and economic development is not stable over time. In earlier periods, countries with intermediate levels of individualism exhibit higher income levels, whereas in later periods, more individualistic countries perform better. This time variation further

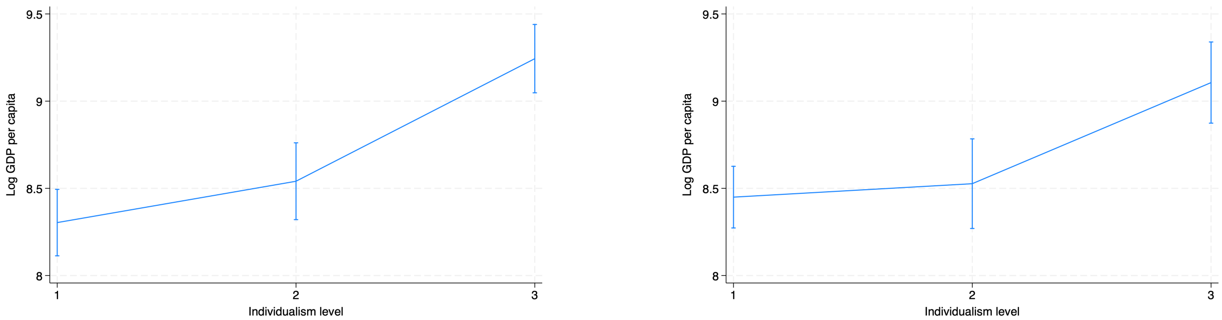
Table 3: Individualism and Patents across countries

	Log Patent per capita (1980-2021)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: OLS Regression						
Individualism	0.936***	0.776***	0.896***	0.895***	0.876***	0.649***
	(0.20)	(0.23)	(0.19)	(0.19)	(0.20)	(0.19)
Adjusted-R^2	0.56	0.63	0.66	0.60	0.58	0.73
Observations	2584	2584	2584	2584	2584	2584
Panel B: IV = DA						
Individualism	1.663***	1.981**	1.374	1.506***	1.539***	1.335
	(0.48)	(0.75)	(0.82)	(0.46)	(0.50)	(1.05)
1st Stage F	9.93	7.53	3.57	9.51	9.33	3.47
Observations	2584	2584	2584	2584	2584	2584
Panel C: Proxy = DA						
Individualism	1.192***	1.082**	0.759	1.074***	1.070**	0.483
	(0.43)	(0.41)	(0.58)	(0.39)	(0.44)	(0.37)
Adjusted-R^2	0.52	0.62	0.61	0.55	0.54	0.71
Observations	2584	2584	2584	2584	2584	2584
Continent FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Continent × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Geo controls	No	Yes	No	No	No	Yes
Legal Origins	No	No	Yes	No	No	Yes
Property Rights	No	No	No	No	Yes	Yes
Religions	No	No	No	Yes	No	Yes

Note: Panels A and B use the standardized individualism measure as the main explanatory variable. Panel B reports 2SLS estimates using DA as an instrument for individualism. Panel C replaces the individualism measure with DA directly as a proxy. All specifications include continent, year, and continent × year fixed effects. Column (2) adds geographic controls; column (3) uses legal-origin controls; column (4) uses religion controls; column (5) uses property-rights controls; and column (6) includes all controls. Standard errors are two-way clustered at the country and year levels. The reported first-stage statistic is the weak-identification F-statistic. *$p < 0.10$, ** $p < 0.05$, ***$p < 0.01$.

weakens the interpretation of a stable cross-country influence of individualism.

The binsreg plot and the time-varying influence on total factor productivity are shown in Figures A7 and A9 in the appendix. The binsreg plot and time-varying influence on patenting activity can be found in Figure A8 and Figure A10 in the appendix.



(a) without controls (b) with controls

Figure 1: Binsreg Regression for economic development across countries

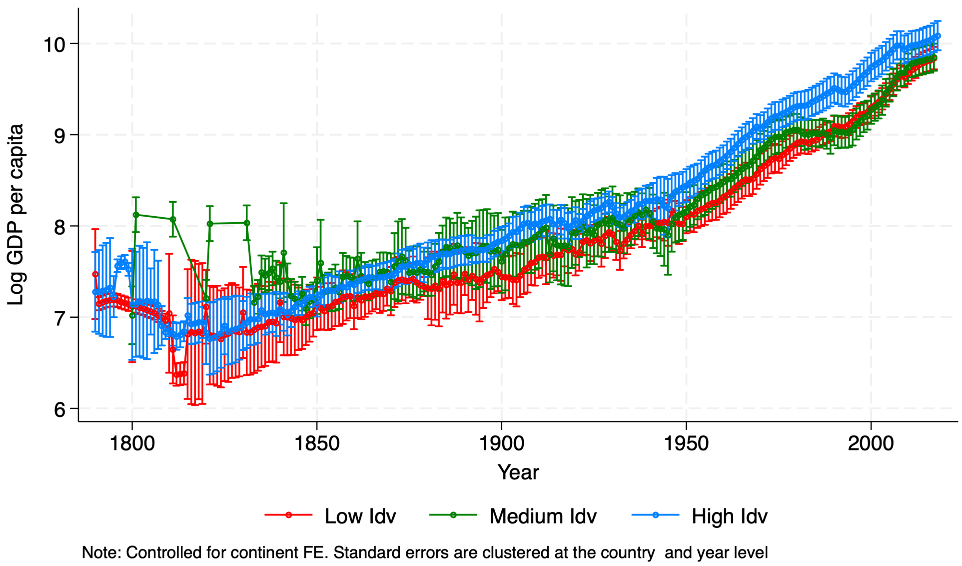


Figure 2: Individualism's impact on economic development across time

3.5 Summary

The cross-country analysis does not provide robust evidence that individualism promotes economic development, productivity, or innovation. Across all outcomes, the estimated relationship is highly sensitive to controls, especially institutional controls, as well as to instruments, proxies, and time periods. These findings suggest that cross-country regressions based on static cultural measures cannot disentangle culture's influence from that of institutions. This motivates the within-U.S. analysis in the next section, where time-varying cultural measurement and local fixed effects enable cleaner identification of the role of individualism.

4 Within-U.S. Analysis

This section reassesses the relationship between individualism and economic outcomes within the United States. The within-country setting provides a useful environment because cultural variation can be studied under a common national institutional context. In addition, the availability of time-varying local measures of individualism enables the inclusion of fixed effects, which helps address concerns about omitted-variable bias that arise in cross-country analyses.

The goal of this section is to examine whether the positive relationship between individualism and economic development documented in the cross-country literature remains robust once institutional differences are better controlled for and cultural variation is measured over time within a single country. Beyond the baseline linear specifications, I also examine whether the relationships between individualism and economic outcomes are nonlinear and whether their influence changes across periods.

Following Fulford et al. (2020), I use the county group as the geographical unit of study since the Census reports data only at this level starting in 1950.

4.1 Measurement of Local Individualism

To construct a time-varying measure of individualism within the United States, I combine country-level individualism indices with county-group ancestry composition. The idea is that ancestry composition captures the cultural traits brought by immigrants and allows for variation of individualism across locations and over time.

Formally, individualism in the county group j at time t is defined as a weighted average of the individualism score at country i:

$$\text{Individualism}_{jt} = \sum_i \text{Ancestry Ratio}_{ijt} \times \text{Individualism}_i. \tag{2}$$

This measure varies across county groups because ancestry composition differs across locations, and evolves over time as ancestry shares change. Compared with standard cross-country measures of culture, this time variation makes it possible to include county-group or state fixed effects, which helps control for unobserved local characteristics and institutional factors.

4.2 Empirical Strategy and Instruments

I use panel regressions to study how the ancestry-weighted individualism measures impact local economic outcomes while controlling for fixed effects and population density using the following equation:

$$Y_{jt} = \beta_0 + \beta_1 Idv_{jt} + \boldsymbol{\beta_2 X_{jt}} + \gamma_t + \phi_j + \omega_{state,t} + \epsilon_{jt} \tag{3}$$

The main endogeneity concern is that ancestry composition may be correlated with local economic performance or regional characteristics. In addition, the country-level modern individualism measures used to construct the index may themselves relate to ancestry composition.

To address the endogeneity concerns, I apply instrumental variables that consider exogenous variation in both ancestry composition and country-level individualism. For ancestry shares, I use instruments based on historical settlement patterns and transportation networks from Fulford et al. (2020). For the country-level individualism component, I use historical measures such as language characteristics related to pronoun drop and historical disease prevalence, following the cross-country literature.

In the baseline specification, I include county-group fixed effects, year fixed effects, and state-by-year fixed effects. This specification controls for time-invariant factors at the county-group level and for common characteristics at the national and state-year levels.

To assess the sensitivity of the results to the choice of fixed effects, I also report specifications that replace county-group fixed effects with state fixed effects. These specifications provide less restrictive controls and allow for greater cross-county-group variation, at the cost of potentially weaker control for local unobserved heterogeneity.

4.3 Linear Regression Results

4.3.1 Economic Development

Table 4 reports the results for log GDP per capita. Overall, the within-U.S. evidence does not support a positive influence of individualism on economic development.

In the baseline specifications with county group fixed effects, the estimated effect of individualism is insignificant but becomes negative once instrument variables for transportation are included. I also consider specifications with only state fixed effects. When state fixed effects are used instead, the estimates of individualism's impact become more consistently negative. Across specifications, the results suggest that higher levels of individualism are not associated with higher economic development in the United States and may even be negatively associated with economic development.

Table 4: Individualism and Log GDP per capita within the US

	Log GDP per capita (1850-2010)					
	OLS		IV: Stock	IV: Trans.	IV: Trans. × DA	IV: Trans. × IHD9
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: With County Group FE						
Individualism	0.001	-0.001	0.008	-0.059***	-0.046***	-0.045***
	(0.00)	(0.00)	(0.01)	(0.01)	(0.01)	(0.01)
County Group FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Year × State FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted-R^2	0.97	0.97				
1st Stage F			5.97	50.79	17.32	33.85
Observations	17631	17631	17631	17631	17631	17631
Panel B: Without County Group FE						
Individualism	-0.016***	-0.014***	-0.014***	-0.059***	-0.054**	-0.055***
	(0.00)	(0.00)	(0.00)	(0.02)	(0.02)	(0.02)
Controls	No	Yes	Yes	Yes	Yes	Yes
County Group FE	No	No	No	No	No	No
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Year × State FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted-R^2	0.91	0.91				
1st Stage F			9.17	17.81	14.61	18.56
Observations	17631	17631	17631	17631	17631	17631

Note: The dependent variable is log GDP per capita. The sample covers 1850–2010. Panel A reports specifications with county-group fixed effects, while Panel B reports specifications without county-group fixed effects but with state fixed effects. Columns (1) and (2) are OLS estimates, where column (2) adds the population density as the control. Columns (3)–(6) are IV estimates using, respectively, the stock-based instrument, the transportation-based instrument, the interaction of transportation and DA, and the interaction of transportation and IHD9. The row labeled "1st Stage F" reports the first-stage F-statistic for the corresponding IV specification. For OLS specifications with county-group fixed effects, standard errors are clustered at the county-group level. For all other specifications, standard errors are clustered at the county-group and year levels. *$p < 0.10$, ** $p < 0.05$, ***$p < 0.01$.

4.3.2 Innovation

Table 5 presents the results for patenting activity. The pattern is consistent with the findings for GDP per capita.

In specifications with county-group fixed effects, the estimated relationship between individualism and innovation is generally insignificant and negative. When state fixed effects are used, the estimated coefficients become more consistently negative.

Overall, these results suggest that, within the United States, individualism does not

Table 5: Individualism and Log Patent per capita within the US

	Log Patent per 1000 people (1850-2010)					
	OLS		IV: Stock	IV: Trans.	IV: Trans. × DA	IV: Trans. × IHD9
Panel A: With County Group FE						
Individualism	-0.006	-0.007	-0.168**	-0.085	-0.131*	-0.083
	(0.01)	(0.01)	(0.06)	(0.06)	(0.07)	(0.05)
County Group FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Year × State FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted-R^2	0.83	0.83				
1st Stage F			6.90	14.76	14.19	34.64
Observations	12591	12591	12591	12591	12591	12591
	(1)	(2)	(3)	(4)	(5)	(6)
Panel B: Without County Group FE						
Individualism	-0.045***	-0.046***	-0.071***	-0.180*	-0.191*	-0.166**
	(0.01)	(0.01)	(0.02)	(0.09)	(0.10)	(0.07)
Controls	No	Yes	Yes	Yes	Yes	Yes
County Group FE	No	No	No	No	No	No
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted-R^2	0.37	0.37				
1st Stage F			8.42	6.32	5.31	8.15
Observations	12591	12591	12591	12591	12591	12591

Note: The dependent variable is log patent per 1000 people. The sample covers 1850–2010. Panel A reports specifications with county-group fixed effects, while Panel B reports specifications without county-group fixed effects but with state fixed effects. Columns (1) and (2) are OLS estimates, where column (2) adds population density as the control. Columns (3)–(6) are IV estimates using, respectively, the stock-based instrument, the transportation-based instrument, the interaction of transportation and DA, and the interaction of transportation and IHD9. The row labeled "1st Stage F" reports the first-stage F-statistic for the corresponding IV specification. For OLS specifications with county-group fixed effects, standard errors are clustered at the county-group level. For all other specifications, standard errors are clustered at the county-group and year levels. *$p < 0.10$, ** $p < 0.05$, ***$p < 0.01$.

robustly promote innovation and may be associated with lower levels of patenting activity.

4.4 Nonlinearity and the Distribution of Individualism

I next examine whether the relationship between individualism and economic outcomes is nonlinear.

4.4.1 Economic Development

Quadratic regressions show that the relationship between individualism and GDP per capita follows an inverted-U pattern, with positive effects at low levels of individualism and negative effects at higher levels. However, these patterns are not robust across all specifications, particularly in the IV regressions with state fixed effects, except when using the instrument of ancestry stock.

To help interpret the difference between the linear and quadratic specifications, Figure 3 plots the fitted relationship between individualism and GDP per capita together with the distribution of county-group individualism. The relationship between individualism and income is U-inverted. The figure also shows that most county groups are concentrated in the upper range of the individualism distribution. Within this region, the fitted relationship is downward sloping. As a result, linear specifications primarily capture the average slope over the high-individualism region, which helps explain why the linear regressions yield negative coefficients even when the quadratic specification suggests a hump-shaped relationship.

4.4.2 Innovation

The results for patenting activity show a similar pattern. While some quadratic specifications suggest a hamp-shaped relationship, the estimates are unstable across specifications and lose significance in most IV regressions.

Figure 4 provides further insight by plotting the fitted relationship between individualism and patenting alongside the distribution of individualism. Similar to the case of GDP, most county groups are located in the higher range of the individualism distribution, where the fitted relationship between individualism and innovation is downward sloping. This pattern helps explain why the linear specifications tend to produce negative estimates for innovation, despite some evidence of nonlinearity in the quadratic regressions.

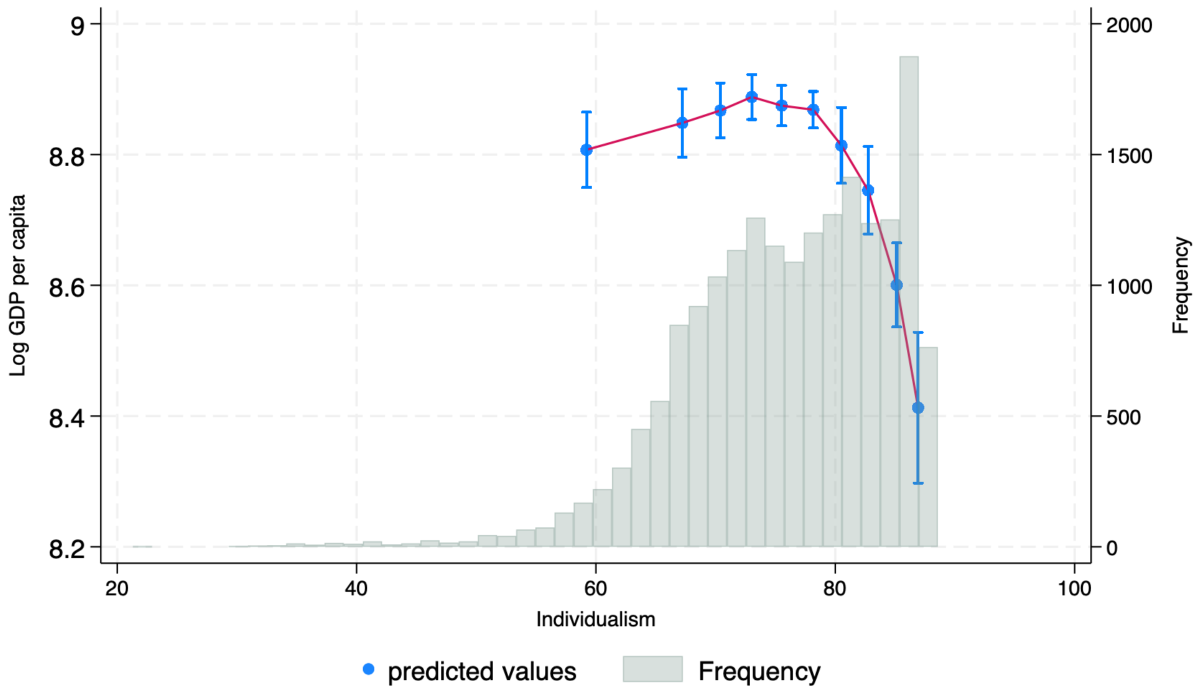


Figure 3: Individualism's impact on economic development within the US

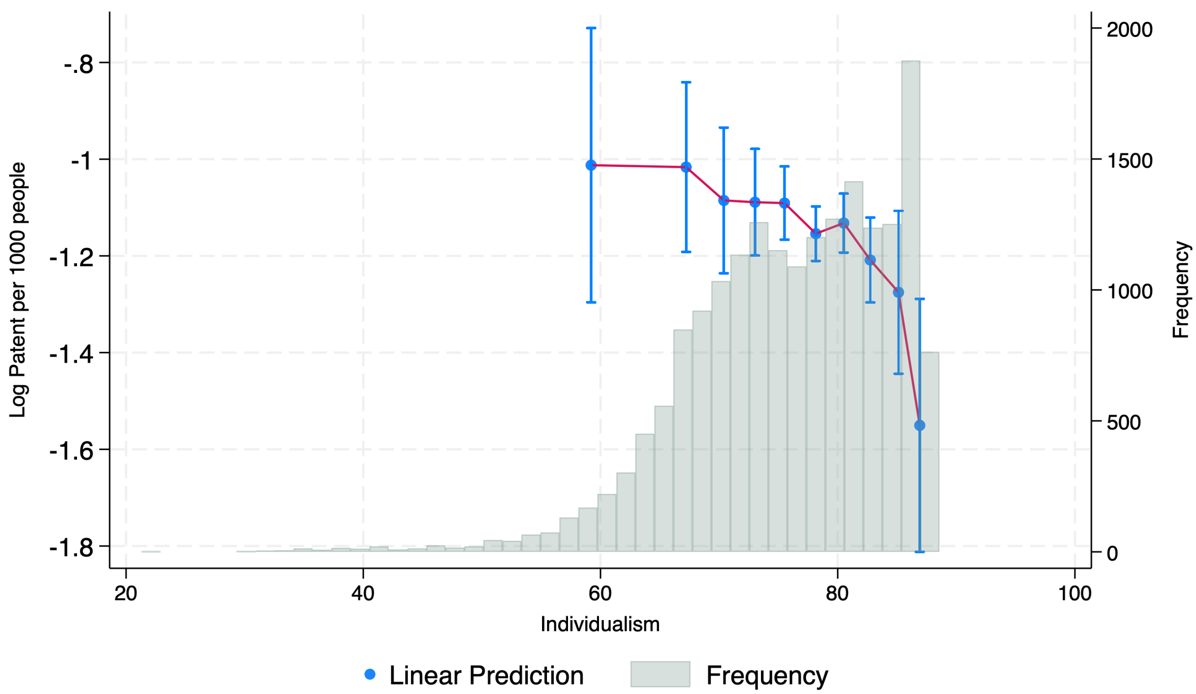


Figure 4: Individualism's impact on patents within the US

Table 6: Individualism and Log GDP per capita within the US

	Log GDP per capita (1850-2010)					
	OLS		IV: Stock	IV: Trans.	IV: Trans. × DA	IV: Trans. × IHD9
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: With County Group FE						
Individualism	0.085***	0.078***	0.190*	0.237***	0.384***	0.284***
	(0.01)	(0.02)	(0.11)	(0.05)	(0.08)	(0.05)
Individualism Square	-0.001***	-0.001***	-0.001*	-0.002***	-0.003***	-0.002***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Turning Point	66.52	65.76	67.51	63.66	67.60	67.30
County Group FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Year × State FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted-R^2	0.97	0.97				
1st Stage F			2.43	1.17	2.91	17.42
Observations	17631	17631	17631	17631	17631	17631
Panel B: Without County Group FE						
Individualism	0.188***	0.195***	0.249***	0.088	-0.375	2.763
	(0.02)	(0.02)	(0.04)	(0.82)	(1.73)	(25.61)
Individualism Square	-0.002***	-0.002***	-0.002***	-0.001	0.002	-0.020
	(0.00)	(0.00)	(0.00)	(0.01)	(0.01)	(0.18)
Controls	No	Yes	Yes	Yes	No	Yes
Turning Point	62.10	62.92	64.01	44.12	82.47	70.84
County Group FE	No	No	No	No	No	No
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted-R^2	0.93	0.93				
1st Stage F			2.23	0.18	0.11	0.00
Observations	17631	17631	17631	17631	17631	17631

Note: The dependent variable is log GDP per capita. The sample covers 1850–2010. Panels A and B report quadratic specifications with two different fixed effects. Panel A reports specifications with county-group fixed effects, while Panel B reports specifications without county-group fixed effects but with state fixed effects. Columns (1) and (2) are OLS estimates, where column (2) adds population density as the control. Columns (3)–(6) are IV estimates using, respectively, the stock-based instrument, the transportation-based instrument, the interaction of transportation and DA, and the interaction of transportation and IHD9. The row labeled "Turning Point" reports the implied turning point from the quadratic specification. The row labeled "1st Stage F" reports the first-stage F-statistic for the corresponding IV specification. Standard errors are clustered at the county-group and year levels. *$p < 0.10$, ** $p < 0.05$, ***$p < 0.01$.

4.5 Time Variation

4.5.1 Economic Development

I next examine whether the relationship between individualism and economic development varies over time. The results indicate substantial time variation. In earlier periods, lower

Table 7: Individualism and Log Patent per capita within the US

	Log Patent per 1000 people (1850-2010)					
	OLS		IV: Stock	IV: Trans.	IV: Trans. × DA	IV: Trans. × IHD9
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: With County Group FE						
Individualism	0.126***	0.121***	-2.267**	0.238	0.648	0.320
	(0.03)	(0.03)	(1.05)	(0.19)	(0.50)	(0.30)
Individualism Square	-0.001***	-0.001***	0.016**	-0.002	-0.005	-0.003
	(0.00)	(0.00)	(0.01)	(0.00)	(0.00)	(0.00)
Turning Point	64.22	63.88	71.52	56.72	61.50	59.62
County Group FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Year × State FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted-R^2	0.83	0.83				
1st Stage F			2.88	0.49	0.76	2.88
Observations	12591	12591	12591	12591	12591	12591
Panel B: Without County Group FE						
Individualism	0.421***	0.423***	-0.180	-0.561	-1.578	-6.135
	(0.07)	(0.07)	(0.36)	(2.50)	(3.88)	(30.86)
Individualism Square	-0.003***	-0.003***	0.001	0.003	0.010	0.042
	(0.00)	(0.00)	(0.00)	(0.02)	(0.03)	(0.22)
Controls	No	Yes	Yes	Yes	Yes	Yes
TurningPoint	60.72	60.82	111.11	109.60	80.84	73.85
County Group FE	No	No	No	No	No	No
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted-R^2	0.41	0.41				
1st Stage F			2.27	0.62	0.60	0.03
Observations	12591	12591	12591	12591	12591	12591

Note: The dependent variable is log patent per 1000 people. The sample covers 1850–2010. Panels A and B report quadratic specifications with two different fixed effects. Panel A reports specifications with county-group fixed effects, while Panel B reports specifications without county-group fixed effects but with state fixed effects. Columns (1) and (2) are OLS estimates, where column (2) adds population density as the control. Columns (3)–(6) are IV estimates using, respectively, the stock-based instrument, the transportation-based instrument, the interaction of transportation and DA, and the interaction of transportation and IHD9. The row labeled "Turning Point" reports the implied turning point from the estimated quadratic specification. The row labeled "1st Stage F" reports the first-stage F-statistic for the corresponding IV specification. Standard errors are clustered at the county-group and year levels. *$p < 0.10$, ** $p < 0.05$, ***$p < 0.01$.

levels of individualism are associated with higher economic performance, while in later periods, higher individualism tends to be associated with better outcomes. This pattern suggests that the effect of individualism is not stable over time.

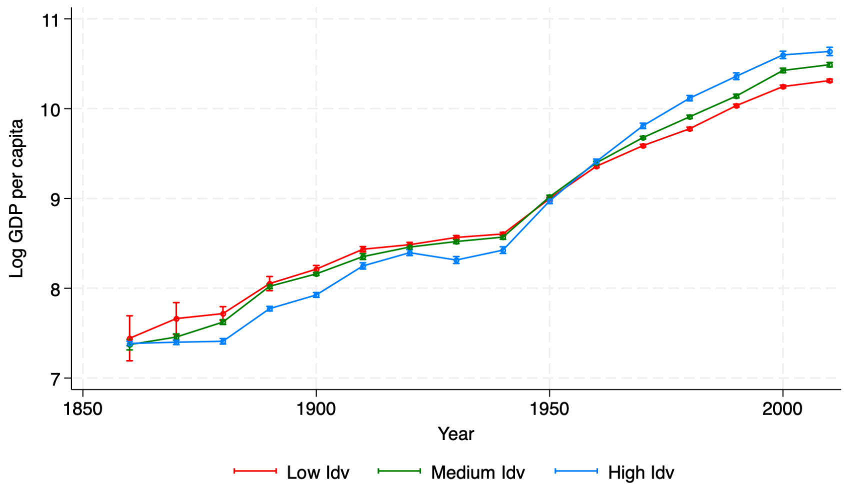


Figure 5: Individualism's Influence on Economic Development within the US over time

4.5.2 Innovation

A similar pattern of time variation is observed for innovation. The relationship between individualism and patenting varies across periods, with no consistently positive or negative association. These results further support the view that the impact of individualism is not constant.

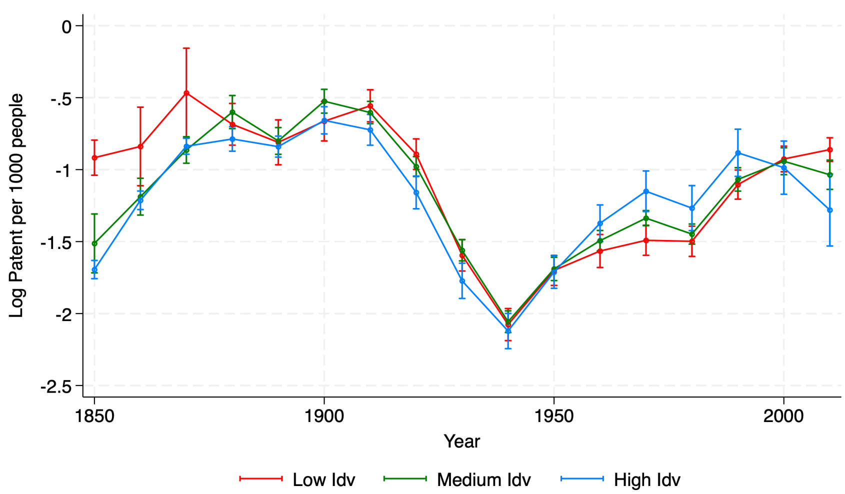


Figure 6: Individualism's Influence on Patents within the US over time

Additional figures and robustness checks, including results with fixed individualism categories, are reported in the appendix.

5 Conclusion

This paper reassesses the impact of individualism on innovation and economic development both across countries and within the US. I begin by revisiting the cross-country evidence that has been interpreted as supporting a positive role for individualism. Extending the analysis over a longer period and introducing richer controls, I show that the estimated relationship is highly sensitive to specification and unstable across time. These patterns suggest that cross-country estimates based on time-invariant cultural measures are difficult to interpret, particularly because they may confound culture with institutions and other omitted variables.

I then turn to a within-U.S. analysis that constructs time-varying ancestry-weighted measures of individualism across county groups. This design allows the inclusion of local fixed effects and provides a setting for studying cultural variation within a common national institutional environment. The within-U.S. results do not support the positive relationship suggested by the cross-country literature. For both economic development and innovation, the estimated effect of individualism is either negative or statistically insignificant once local fixed effects are accounted for and instruments are used. Additional evidence of nonlinearity and time variation further suggests that the impact of individualism is unstable and cannot be characterized as universally positive.

These findings should be interpreted with appropriate caution. Although the within-U.S. analysis helps address some of the limitations of the cross-country approach, the results may still be specific to the U.S. setting. In particular, the effect of individualism may depend on the institutional and social environment. One possible interpretation is that in a highly individualistic society such as the United States, additional individualism may weaken cooperation rather than strengthen innovation and economic performance. Whether similar patterns arise in other countries remains an open question.

More broadly, the results suggest that the economic effects of culture are likely to be context-dependent rather than universal. Future research could extend this analysis to other countries with time-varying measures of culture and examine how cultural traits interact with institutions and other features of the social environment. In addition, this paper focuses on the quantity of innovation rather than its quality. Studying how individualism affects the quality of innovation would provide a more comprehensive understanding of culture's role in technological progress and economic development.

References

Alesina, A. and Giuliano, P. (2014). Family ties, *Handbook of economic growth*, Vol. 2, Elsevier, pp. 177–215.

Andersen, T. B., Bentzen, J., Dalgaard, C.-J. and Sharp, P. (2017). Pre-reformation roots of the protestant ethic, *The Economic Journal* **127**(604): 1756–1793.

Becker, S. O. and Woessmann, L. (2009). Was weber wrong? a human capital theory of protestant economic history, *The quarterly journal of economics* **124**(2): 531–596.

Davis, L. S. and Abdurazokzoda, F. (2016). Language, culture and institutions: Evidence from a new linguistic dataset, *Journal of Comparative Economics* **44**(3): 541–561.

Dohmen, T., Enke, B., Falk, A., Huffman, D., Sunde, U. et al. (2015). Patience and the wealth of nations, *Unpublished manuscript. University of Bonn* .

Fulford, S. L., Petkov, I. and Schiantarelli, F. (2020). Does it matter where you came from? ancestry composition and economic performance of us counties, 1850–2010, *Journal of Economic Growth* **25**(3): 341–380.

Galor, O. and Özak, Ö. (2016). The agricultural origins of time preference, *American Economic Review* **106**(10): 3064–3103.

Giavazzi, F., Petkov, I. and Schiantarelli, F. (2019). Culture: Persistence and evolution, *Journal of Economic Growth* **24**: 117–154.

Gorodnichenko, Y. and Roland, G. (2011a). Individualism, innovation, and long-run growth, *Proceedings of the National Academy of Sciences* **108**(Supplement 4): 21316–21319.

Gorodnichenko, Y. and Roland, G. (2011b). Which dimensions of culture matter for long-run growth?, *American Economic Review* **101**(3): 492–98.

Gorodnichenko, Y. and Roland, G. (2017). Culture, institutions, and the wealth of nations, *Review of Economics and Statistics* **99**(3): 402–416.

Hofstede, G., Hofstede, G. J. and Minkov, M. (2005). *Cultures and organizations: Software of the mind*, Vol. 2, Mcgraw-hill New York.

Kashima, E. S. and Kashima, Y. (1998). Culture and language: The case of cultural dimensionsand personal pronoun use, *Journal of cross-cultural psychology* **29**(3): 461–486.

Murray, D. R. and Schaller, M. (2010). Historical prevalence of infectious diseases within 230 geopolitical regions: A tool for investigating origins of culture, *Journal of Cross-Cultural Psychology* **41**(1): 99–108.

Sequeira, S., Nunn, N. and Qian, N. (2020). Immigrants and the making of america, *The Review of Economic Studies* **87**(1): 382–419.

Zak, P. J. and Knack, S. (2001). Trust and growth, *The economic journal* **111**(470): 295–321.

Appendix

A Additional Cross-country Results

A.1 Cross-country Results Across Specifications

This section reports additional country-level results not shown in the main text. For each outcome, I present alternative IV and proxy specifications using historical measures of individualism based on pronoun-drop language characteristics and historical disease prevalence. Across these alternative specifications, the estimated relationship between individualism and economic outcomes remains sensitive to controls, instruments, and measurement .

A.1.1 Log GDP per capita

Table A1 shows the regression results for individualism's impact on economic development across countries using the pronoun drop from Kashima and Kashima (1998) as the instrument or proxy. The results in Panel A are the same as those in the paper, and I put them here just for reference. When I use the pronoun drop as the instrument, it can be noticed that the coefficient drops when I include legal origins and property rights. When I consider geographic controls and religions, the coefficient increases. These inconsistent results indicate that individualism could correlate with different controls, leading to biased results in the country-level analysis. In Panel C, when we use the pronoun drop as the proxy, we see similar results as those in Panel B.

Table A2 and Table A3 show individualism's influence on economic development across

Table A1: Individualism and Economic Development across countries

	Log GDP per capita (1789-2018)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: OLS Regression						
Individualism	0.453***	0.411***	0.327***	0.458***	0.392***	0.267**
	(0.10)	(0.13)	(0.08)	(0.11)	(0.11)	(0.10)
Adjusted-R^2	0.80	0.80	0.84	0.80	0.81	0.85
Observations	5938	5938	5938	5938	5938	5938
Panel B: IV = KK						
Individualism	0.767***	1.020***	0.482***	0.779***	0.724***	1.136*
	(0.16)	(0.27)	(0.13)	(0.16)	(0.18)	(0.65)
1st Stage F	16.26	7.43	10.53	16.04	10.49	2.13
Observations	5938	5938	5938	5938	5938	5938
Panel C: Proxy = KK						
Individualism	0.815***	0.843***	0.483***	0.823***	0.671***	0.541***
	(0.17)	(0.18)	(0.13)	(0.17)	(0.18)	(0.15)
Adjusted-R^2	0.78	0.84	0.82	0.78	0.79	0.87
Observations	5938	5938	5938	5938	5938	5938
Continent FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Continent × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Geo controls	No	Yes	No	No	No	Yes
Legal Origins	No	No	Yes	No	No	Yes
Property Rights	No	No	No	No	Yes	Yes
Religions	No	No	No	Yes	No	Yes

Note: The dependent variable is log GDP per capita. The sample covers 1789–2018. Panel A reports OLS estimates using the standardized individualism measure as the main explanatory variable. Panel B reports 2SLS estimates using KK as an instrument for individualism. Panel C replaces the individualism measure with KK directly as a proxy. Columns (2)–(5) add, respectively, geographic controls, legal-origin controls, religion controls, and property-rights controls, while column (6) includes the full set of controls. The row labeled "1st Stage F" reports the first-stage F-statistic for the corresponding IV specification. The specification rows reported at the bottom apply to all three panels. Standard errors are clustered at the country and year levels. *$p < 0.10$, ** $p < 0.05$, ***$p < 0.01$..

countries using the indexes of 9 and 7 historical diseases, respectively. In Table A2, the IV regression results show that the influence of individualism vanishes or drops when I consider geographic controls and legal origins. When I consider property rights and religions, the coefficient increases. In Panel C, when I use the index of 9 historical diseases as the proxy, it can be noticed that the influence of individualism drops in every column when adding controls. In Table A3, wI apply the index of 7 historical diseases as the instrument and the proxy. In Panel B, the coefficient decreases when I include legal origins. In Panel C, there is a drop in the influence when I add controls.

A.1.2 Total Factor Productivity

Table A4 indicates the regression results based on the pronoun drop from Kashima and Kashima (1998) for TFP. When I use the pronoun drop as the instrument, the significance of the coefficients disappears for most of the controls, except for religions. When I use the pronoun drop as the proxy instead, all the significance disappears.

Table A5 lists the regression results for TFP using the index of 9 historical diseases as the instrument and the proxy. The magnitude or significance of coefficients in both panels drops, except for the case when considering legal origins. This indicates that individualism might be highly correlated with legal origins. Similarly, Table A6 shows the influence of individualism on TFP when I use the index of 7 historical diseases as the instrument and the proxy. There exists a similar pattern for those in Table A5 .

A.1.3 Patents

Finally, I check the influence of individualism on innovation using different instruments and proxies. In Table A7, I use the pronoun drop from Kashima and Kashima (1998) as the instrument and the proxy. For both panels, except when I include geographic characteristics as the control, the magnitude or the significance of individualism's impact decreases. In

Table A2: Individualism and Economic Development across countries

	Log GDP per capita (1789-2018)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: OLS Regression						
Individualism	0.496***	0.446***	0.348***	0.496***	0.423***	0.257***
	(0.08)	(0.10)	(0.07)	(0.08)	(0.09)	(0.09)
Adjusted-R^2	0.74	0.75	0.79	0.75	0.76	0.81
Observations	7123	7123	7123	7123	7123	7123
Panel B: IV = Index of 9 diseases						
Individualism	1.176**	3.360	1.089**	1.209**	1.215**	5.685
	(0.46)	(5.27)	(0.52)	(0.51)	(0.54)	(15.27)
1st Stage F	5.30	0.35	3.98	4.78	4.75	0.13
Observations	7123	7123	7123	7123	7123	7123
Panel C: Proxy = Index of 9 diseases						
Individualism	0.455***	0.378***	0.380***	0.447***	0.410***	0.382***
	(0.09)	(0.13)	(0.07)	(0.10)	(0.09)	(0.11)
Adjusted-R^2	0.70	0.75	0.79	0.70	0.74	0.83
Observations	7123	7123	7123	7123	7123	7123
Continent FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Continent × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Geo controls	No	Yes	No	No	No	Yes
Legal Origins	No	No	Yes	No	No	Yes
Property Rights	No	No	No	No	Yes	Yes
Religions	No	No	No	Yes	No	Yes

Note: The dependent variable is log GDP per capita. The sample covers 1789–2018. Panel A reports OLS estimates using the standardized individualism measure as the main explanatory variable. Panel B reports 2SLS estimates using the index of nine diseases as an instrument for individualism. Panel C replaces the individualism measure with the index of nine diseases directly as a proxy. Columns (2)–(5) add, respectively, geographic controls, legal-origin controls, religion controls, and property-rights controls, while column (6) includes the full set of controls. The row labeled "1st Stage F" reports the first-stage F-statistic for the corresponding IV specification. The specification rows reported at the bottom apply to all three panels. Standard errors are clustered at the country and year levels. *$p < 0.10$, ** $p < 0.05$, ***$p < 0.01$.

Table A8, I use the index of nine historical diseases. For Panel B and Panel C, the magnitude and significance drop except for the case when including religions. Table A9 shows the results using the index of 7 historical diseases. We see similar results to those in Table A8.

Table A3: Individualism and Economic Development across countries

	Log GDP per capita (1789-2018)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: OLS Regression						
Individualism	0.493***	0.437***	0.350***	0.492***	0.419***	0.244***
	(0.08)	(0.10)	(0.07)	(0.08)	(0.09)	(0.09)
Adjusted-R^2	0.74	0.75	0.79	0.74	0.76	0.80
Observations	7171	7171	7171	7171	7171	7171
Panel B: IV = Index of 7 diseases						
Individualism	1.153*	5.637	1.046*	1.197*	1.187*	-13.205
	(0.60)	(22.91)	(0.59)	(0.67)	(0.71)	(128.89)
1st Stage F	3.09	0.05	2.82	2.82	2.69	0.01
Observations	7171	7171	7171	7171	7171	7171
Panel C: Proxy = Index of 7 diseases						
Individualism	0.307***	0.227*	0.280***	0.296***	0.274***	0.216*
	(0.09)	(0.13)	(0.08)	(0.09)	(0.09)	(0.11)
Adjusted-R^2	0.67	0.74	0.77	0.68	0.72	0.82
Observations	7171	7171	7171	7171	7171	7171
Continent FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Continent × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Geo controls	No	Yes	No	No	No	Yes
Legal Origins	No	No	Yes	No	No	Yes
Property Rights	No	No	No	No	Yes	Yes
Religions	No	No	No	Yes	No	Yes

Note: The dependent variable is log GDP per capita. The sample covers 1789–2018. Panel A reports OLS estimates using the standardized individualism measure as the main explanatory variable. Panel B reports 2SLS estimates using the index of seven diseases as an instrument for individualism. Panel C replaces the individualism measure with the index of seven diseases directly as a proxy. Columns (2)–(5) add, respectively, geographic controls, legal-origin controls, religion controls, and property-rights controls, while column (6) includes the full set of controls. The row labeled "1st Stage F" reports the first-stage F-statistic for the corresponding IV specification. The specification rows reported at the bottom apply to all three panels. Standard errors are clustered at the country and year levels. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A4: Individualism and Economic Development across countries

	Total Factor Productivity (1954-2019)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: OLS Regression						
Individualism	0.081***	0.014	0.080**	0.078**	0.072**	0.025
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Adjusted-R^2	0.25	0.37	0.34	0.35	0.26	0.45
Observations	3179	3179	3179	3179	3179	3179
Panel B: IV = KK						
Individualism	0.070	0.026	0.079	0.089*	0.054	0.114*
	(0.06)	(0.06)	(0.06)	(0.05)	(0.06)	(0.06)
1st Stage F	18.44	11.19	15.15	18.18	16.48	8.53
Observations	3179	3179	3179	3179	3179	3179
Panel C: Proxy = KK						
Individualism	0.081	0.027	0.092	0.103	0.061	0.101
	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.06)
Adjusted-R^2	0.21	0.50	0.30	0.32	0.23	0.58
Observations	3179	3179	3179	3179	3179	3179
Continent FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Continent × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Geo controls	No	Yes	No	No	No	Yes
Legal Origins	No	No	Yes	No	No	Yes
Institutions	No	No	No	No	Yes	Yes
Religions	No	No	No	Yes	No	Yes

Note: The dependent variable is total factor productivity. The sample covers 1954–2019. Panel A reports OLS estimates using the standardized individualism measure as the main explanatory variable. Panel B reports 2SLS estimates using KK as an instrument for individualism. Panel C replaces the individualism measure with KK directly as a proxy. Columns (2)–(5) add, respectively, geographic controls, legal-origin controls, religion controls, and property-rights controls, while column (6) includes the full set of controls. The row labeled "1st Stage F" reports the first-stage F-statistic for the corresponding IV specification. The specification rows reported at the bottom apply to all three panels. Standard errors are clustered at the country and year levels. *$p < 0.10$, ** $p < 0.05$, ***$p < 0.01$.

Table A5: Individualism and Economic Development across countries

	Total Factor Productivity (1954-2019)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: OLS Regression						
Individualism	0.086***	0.030	0.060*	0.081***	0.076**	0.001
	(0.03)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)
Adjusted-R^2	0.17	0.27	0.28	0.28	0.19	0.38
Observations	3776	3776	3776	3776	3776	3776
Panel B: IV = Index of 9 diseases						
Individualism	0.293**	0.530	0.324**	0.253**	0.289**	0.900
	(0.12)	(0.44)	(0.13)	(0.11)	(0.13)	(1.18)
1st Stage F	7.92	1.69	6.78	7.67	7.13	0.62
Observations	3776	3776	3776	3776	3776	3776
Panel C: Proxy = Index of 9 diseases						
Individualism	0.129***	0.127***	0.138***	0.113***	0.123***	0.139***
	(0.03)	(0.04)	(0.03)	(0.03)	(0.03)	(0.04)
Adjusted-R^2	0.21	0.41	0.35	0.30	0.24	0.47
Observations	3776	3776	3776	3776	3776	3776
Continent FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Continent $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Geo controls	No	Yes	No	No	No	Yes
Legal Origins	No	No	Yes	No	No	Yes
Property Rights	No	No	No	No	Yes	Yes
Religions	No	No	No	Yes	No	Yes

Note: The dependent variable is total factor productivity. The sample covers 1954–2019. Panel A reports OLS estimates using the standardized individualism measure as the main explanatory variable. Panel B reports 2SLS estimates using the index of nine diseases as an instrument for individualism. Panel C replaces the individualism measure with the index of nine diseases directly as a proxy. Columns (2)–(5) add, respectively, geographic controls, legal-origin controls, religion controls, and property-rights controls, while column (6) includes the full set of controls. The row labeled "1st Stage F" reports the first-stage F-statistic for the corresponding IV specification. The specification rows reported at the bottom apply to all three panels. Standard errors are clustered at the country and year levels. *$p < 0.10$, ** $p < 0.05$, ***$p < 0.01$.

A.2 Graphical Evidence on Specification Sensitivity

Figures A7 and Figure A8 present additional binned scatter plots for TFP and patents. The relationship between individualism and TFP becomes a U-shape once controls are included.

Table A6: Individualism and Economic Development across countries

	Total Factor Productivity (1954-2019)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: OLS Regression						
Individualism	0.087***	0.037	0.060*	0.082***	0.078**	0.004
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Adjusted-R^2	0.18	0.27	0.29	0.28	0.20	0.39
Observations	3822	3822	3822	3822	3822	3822
Panel B: IV = Index of 7 diseases						
Individualism	0.185*	0.358	0.246**	0.174*	0.178	0.820
	(0.11)	(0.40)	(0.12)	(0.10)	(0.11)	(1.73)
1st Stage F	4.99	0.95	5.33	4.95	4.48	0.24
Observations	3822	3822	3822	3822	3822	3822
Panel C: Proxy = Index of 7 diseases						
Individualism	0.059**	0.059*	0.086***	0.056**	0.055**	0.069
	(0.03)	(0.04)	(0.02)	(0.03)	(0.03)	(0.04)
Adjusted-R^2	0.15	0.37	0.31	0.26	0.18	0.44
Observations	3822	3822	3822	3822	3822	3822
Continent FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Continent × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Geo controls	No	Yes	No	No	No	Yes
Legal Origins	No	No	Yes	No	No	Yes
Property Rights	No	No	No	No	Yes	Yes
Religions	No	No	No	Yes	No	Yes

Note: The dependent variable is total factor productivity. The sample covers 1954–2019. Panel A reports OLS estimates using the standardized individualism measure as the main explanatory variable. Panel B reports 2SLS estimates using the index of seven diseases as an instrument for individualism. Panel C replaces the individualism measure with the index of seven diseases directly as a proxy. Columns (2)–(5) add, respectively, geographic controls, legal-origin controls, religion controls, and property-rights controls, while column (6) includes the full set of controls. The row labeled "1st Stage F" reports the first-stage F-statistic for the corresponding IV specification. The specification rows reported at the bottom apply to all three panels. Standard errors are clustered at the country and year levels. *$p < 0.10$, ** $p < 0.05$, ***$p < 0.01$.

The results for patents are similar to those for log GDP per capita and become flatter when I consider control variables.

Table A7: Individualism and Patents across countries

	Log Patent per capita (1980-2021)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: OLS Regression						
Individualism	0.957***	0.791***	1.003***	0.895***	0.899***	0.844***
	(0.24)	(0.28)	(0.23)	(0.21)	(0.24)	(0.20)
Adjusted-R^2	0.57	0.66	0.69	0.64	0.58	0.78
Observations	2137	2137	2137	2137	2137	2137
Panel B: IV = KK						
Individualism	1.145***	1.450***	0.658*	1.049***	1.067***	0.894**
	(0.24)	(0.36)	(0.33)	(0.24)	(0.26)	(0.41)
1st Stage F	37.50	21.18	35.09	36.45	35.28	12.57
Observations	2137	2137	2137	2137	2137	2137
Panel C: Proxy = KK						
kkquants	1.529***	1.621***	0.932*	1.403***	1.395***	0.925*
	(0.32)	(0.42)	(0.49)	(0.33)	(0.35)	(0.49)
Adjusted-R^2	0.53	0.67	0.62	0.60	0.54	0.76
Observations	2137	2137	2137	2137	2137	2137
Continent FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Continent $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Geo controls	No	Yes	No	No	No	Yes
Legal Origins	No	No	Yes	No	No	Yes
Property Rights	No	No	No	No	Yes	Yes
Religions	No	No	No	Yes	No	Yes

Note: The dependent variable is log patent per capita. The sample covers 1980–2021. Panel A reports OLS estimates using the standardized individualism measure as the main explanatory variable. Panel B reports 2SLS estimates using KK as an instrument for individualism. Panel C replaces the individualism measure with KK directly as a proxy. Columns (2)–(5) add, respectively, geographic controls, legal-origin controls, religion controls, and property-rights controls, while column (6) includes the full set of controls. The row labeled "1st Stage F" reports the first-stage F-statistic for the corresponding IV specification. The specification rows reported at the bottom apply to all three panels. Standard errors are clustered at the country and year levels. *$p < 0.10$, ** $p < 0.05$, ***$p < 0.01$.

A.3 Additional Evidence on Time Variation

Figure A9 and Figure A10 show the result of time-varying estimates for TFP and patents. It has been shown that individualism's influence is not time-consistent.

Table A8: Individualism and Patents across countries

	Log Patent per capita (1980-2021)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: OLS Regression						
Individualism	0.895***	0.779***	0.899***	0.853***	0.842***	0.583***
	(0.19)	(0.20)	(0.20)	(0.17)	(0.20)	(0.18)
Adjusted-R^2	0.54	0.60	0.64	0.58	0.55	0.71
Observations	2838	2838	2838	2838	2838	2838
Panel B: IV = Index of 9 diseases						
Individualism	1.715***	1.648	1.071**	1.871***	1.679***	1.343
	(0.59)	(1.09)	(0.42)	(0.60)	(0.59)	(0.80)
1st Stage F	12.85	3.14	9.74	13.81	11.62	2.94
Observations	2838	2838	2838	2838	2838	2838
Panel C: Proxy = Index of 9 diseases						
Individualism	0.898***	0.533	0.527**	1.017***	0.850***	0.434
	(0.27)	(0.39)	(0.24)	(0.24)	(0.26)	(0.30)
Adjusted-R^2	0.50	0.58	0.59	0.57	0.52	0.70
Observations	2838	2838	2838	2838	2838	2838
Continent FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Continent $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Geo controls	No	Yes	No	No	No	Yes
Legal Origins	No	No	Yes	No	No	Yes
Property Rights	No	No	No	No	Yes	Yes
Religions	No	No	No	Yes	No	Yes

Note: The dependent variable is log patent per capita. The sample covers 1980–2021. Panel A reports OLS estimates using the standardized individualism measure as the main explanatory variable. Panel B reports 2SLS estimates using the index of nine diseases as an instrument for individualism. Panel C replaces the individualism measure with the index of nine diseases directly as a proxy. Columns (2)–(5) add, respectively, geographic controls, legal-origin controls, religion controls, and property-rights controls, while column (6) includes the full set of controls. The row labeled "1st Stage F" reports the first-stage F-statistic for the corresponding IV specification. The specification rows reported at the bottom apply to all three panels. Standard errors are clustered at the country and year levels. *$p < 0.10$, ** $p < 0.05$, ***$p < 0.01$.

B Additional Within-U.S. Results

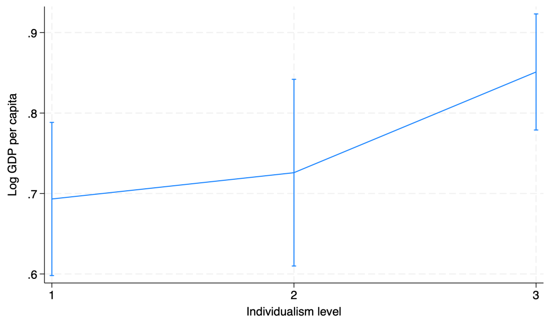
This section reports supplementary results for the within-U.S. analysis. I first present alternative instrumental variable specifications, then report additional evidence on nonlinearity,

Table A9: Individualism and Patents across countries

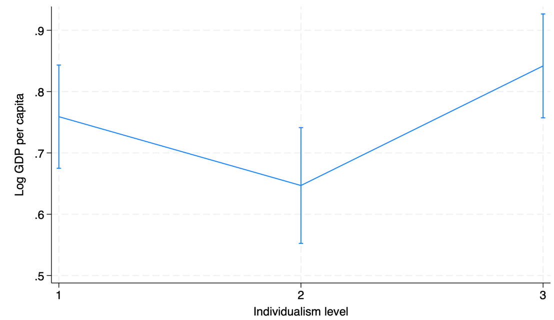
	Log Patent per capita (1980-2021)					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: OLS Regression						
Individualism	0.871***	0.730***	0.894***	0.828***	0.815***	0.531***
	(0.19)	(0.19)	(0.20)	(0.17)	(0.19)	(0.18)
Adjusted-R^2	0.53	0.60	0.64	0.57	0.54	0.70
Observations	2887	2887	2887	2887	2887	2887
Panel B: IV = Index of 7 diseases						
Individualism	2.215**	1.675	1.174**	2.239**	2.186**	0.988
	(0.93)	(1.54)	(0.56)	(0.90)	(0.93)	(1.17)
1st Stage F	8.04	1.69	7.31	8.31	6.95	1.50
Observations	2887	2887	2887	2887	2887	2887
Panel C: Proxy = Index of 7 diseases						
Individualism	0.816***	0.365	0.453*	0.831***	0.767***	0.204
	(0.26)	(0.39)	(0.25)	(0.23)	(0.26)	(0.30)
Adjusted-R^2	0.50	0.57	0.58	0.55	0.52	0.69
Observations	2887	2887	2887	2887	2887	2887
Continent FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Continent × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Geo controls	No	Yes	No	No	No	Yes
Legal Origins	No	No	Yes	No	No	Yes
Property Rights	No	No	No	No	Yes	Yes
Religions	No	No	No	Yes	No	Yes

Note: The dependent variable is log patent per capita. The sample covers 1980–2021. Panel A reports OLS estimates using the standardized individualism measure as the main explanatory variable. Panel B reports 2SLS estimates using the index of seven diseases as an instrument for individualism. Panel C replaces the individualism measure with the index of seven diseases directly as a proxy. Columns (2)–(5) add, respectively, geographic controls, legal-origin controls, religion controls, and property-rights controls, while column (6) includes the full set of controls. The row labeled "1st Stage F" reports the first-stage F-statistic for the corresponding IV specification. The specification rows reported at the bottom apply to all three panels. Standard errors are clustered at the country and year levels. *$p < 0.10$, ** $p < 0.05$, ***$p < 0.01$.

and finally provide supplementary figures that examine time variation using fixed individualism categories at one particular year. These results complement the main findings in the paper and assess the robustness of the estimated influence.

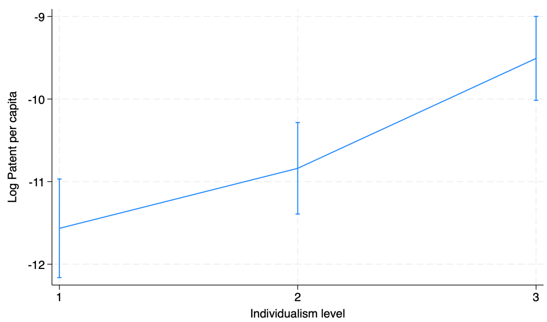


(a) without controls

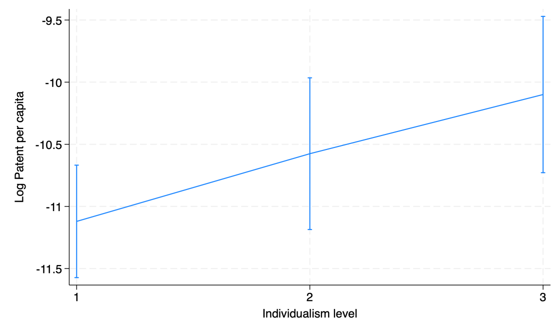


(b) with controls

Figure A7: Binsreg Regression for TFP across countries



(a) without controls



(b) with controls

Figure A8: Binsreg Regression for patents across countries

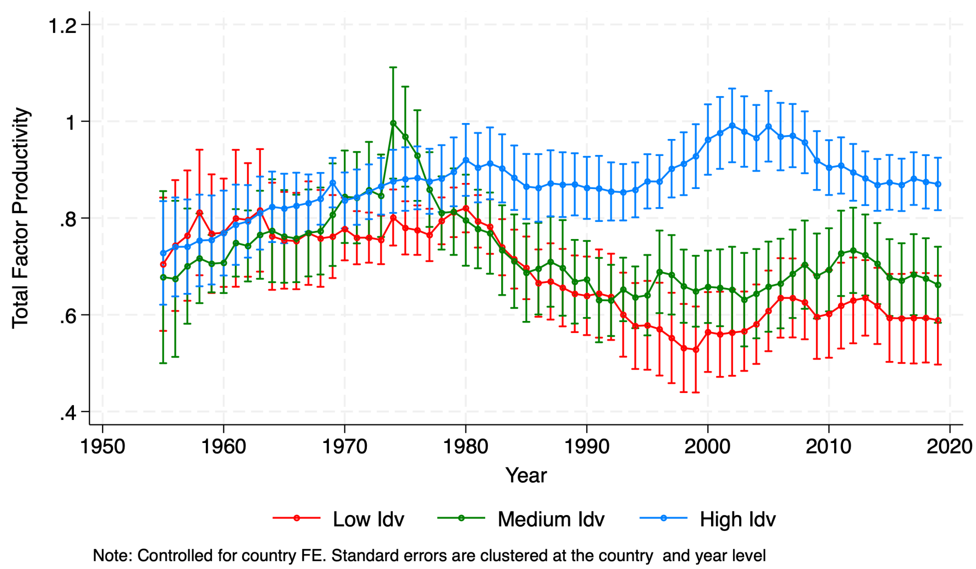


Figure A9: Individualism's impact on TFP across time

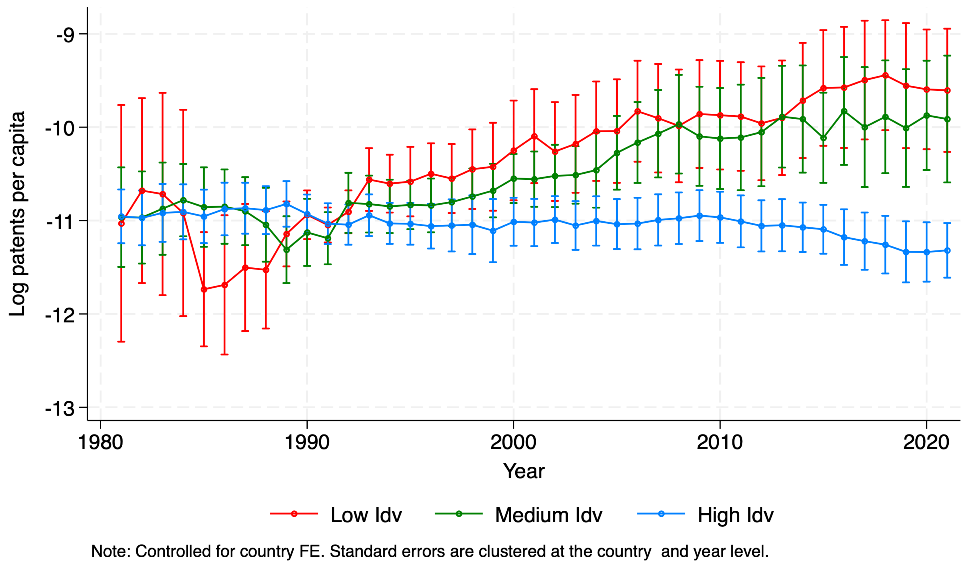


Figure A10: Individualism's impact on patents across time

B.1 Alternative IV Specifications

B.1.1 Economic Development

Table A10 reports the results for log GDP per capita using alternative instruments based on the pronoun-drop measure from Kashima and Kashima (1998) and the index of seven historical diseases from Murray and Schaller (2010).

The overall pattern is consistent with the main results. When county-group fixed effects are included, the estimated effect of individualism is generally statistically insignificant in OLS and in the ancestry-stock IV specification, but becomes negative and significant when using transportation-based instruments. When state fixed effects are included instead, the estimated influence is consistently negative across specifications.

Table A10: Individualism and Log GDP per capita within the US

	Log GDP per capita (1850-2010)					
	OLS		IV: Stock	IV: Trans.	IV: Trans. × KK	IV: Trans. × IHD7
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: With County Group FE						
Individualism	0.001	-0.001	0.008	-0.059***	-0.041***	-0.038***
	(0.00)	(0.00)	(0.01)	(0.01)	(0.01)	(0.01)
Pop Density	No	Yes	Yes	Yes	Yes	Yes
County Group FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Year × State FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted-R^2	0.97	0.97				
1st Stage F			5.97	50.79	17.36	34.49
Observations	17631	17631	17631	17631	17631	17631
Panel B: Without County Group FE						
Individualism	-0.016***	-0.014***	-0.014***	-0.059***	-0.053**	-0.049***
	(0.00)	(0.00)	(0.00)	(0.02)	(0.02)	(0.01)
Pop Density	No	Yes	Yes	Yes	Yes	Yes
County Group FE	No	No	No	No	No	No
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted-R^2	0.91	0.91				
1st Stage F			9.17	17.81	13.10	17.75
Observations	17631	17631	17631	17631	17631	17631

Note: The dependent variable is log GDP per capita. The sample covers 1850–2010. Panel A reports specifications with county-group fixed effects, while Panel B reports specifications without county-group fixed effects but with state fixed effects. Columns (1) and (2) are OLS estimates. Columns (3)–(6) are IV estimates using, respectively, the stock-based instrument, the transportation-based instrument, the interaction of transportation and KK, and the interaction of transportation and IHD7. The row labeled "Pop Density" indicates whether population density is included as a control. The row labeled "1st Stage F" reports the first-stage F-statistic for the corresponding IV specification. For OLS specifications with county-group fixed effects, standard errors are clustered at the county-group level. For all other specifications, standard errors are clustered at the county-group and year levels. $*p < 0.10$, $**\ p < 0.05$, $***p < 0.01$.

B.1.2 Innovation

Table A11 presents the corresponding results for patenting activity. The findings are similar to those for economic development. Across specifications, individualism has either a negative or statistically insignificant effect on innovation.

Table A11: Individualism and Log Patent per capita within the US

	Log Patent per 1000 people (1850-2010)					
	OLS		IV: Stock	IV: Trans.	IV: Trans. × KK	IV: Trans. × IHD7
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: With County Group FE						
Individualism	-0.006	-0.007	-0.168**	-0.085	-0.132*	-0.087*
	(0.01)	(0.01)	(0.06)	(0.06)	(0.07)	(0.05)
Pop Density	No	Yes	Yes	Yes	Yes	Yes
County Group FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Year × State FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted-R^2	0.83	0.83				
1st Stage F			6.90	14.76	13.54	42.69
Observations	12591	12591	12591	12591	12591	12591
	(1)	(2)	(3)	(4)	(5)	(6)
Panel B: Without County Group FE						
Individualism	-0.045***	-0.046***	-0.071***	-0.180*	-0.179	-0.146**
	(0.01)	(0.01)	(0.02)	(0.09)	(0.11)	(0.06)
Pop Density	No	Yes	Yes	Yes	Yes	Yes
County Group FE	No	No	No	No	No	No
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted-R^2	0.37	0.37				
1st Stage F			8.42	6.32	4.53	8.65
Observations	12591	12591	12591	12591	12591	12591

Note: The dependent variable is log patent per 1000 people. The sample covers 1850–2010. Panel A reports specifications with county-group fixed effects, while Panel B reports specifications without county-group fixed effects but with state fixed effects. Columns (1) and (2) are OLS estimates. Columns (3)–(6) are IV estimates using, respectively, the stock-based instrument, the transportation-based instrument, the interaction of transportation and KK, and the interaction of transportation and IHD7. The row labeled "Pop Density" indicates whether population density is included as a control. The row labeled "1st Stage F" reports the first-stage F-statistic for the corresponding IV specification. For OLS specifications with county-group fixed effects, standard errors are clustered at the county-group level. For all other specifications, standard errors are clustered at the county-group and year levels. *$p < 0.10$, ** $p < 0.05$, ***$p < 0.01$.

B.2 Additional Evidence on Nonlinearity

B.2.1 Economic Development

Table A12 reports quadratic specifications for log GDP per capita using the alternative instruments. The results are broadly consistent with those in the paper. Some specifications suggest an inverted-U relationship between individualism and economic development, with positive effects at lower levels of individualism and negative effects at higher levels. However, this nonlinear pattern is not stable across all IV specifications.

Table A12: Individualism and Log GDP per capita within the US

	Log GDP per capita (1850-2010)					
	OLS		IV: Stock	IV: Trans.	IV: Trans. × KK	IV: Trans. × IHD7
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: With County Group FE						
Individualism	0.085***	0.078***	0.190*	0.237***	0.384***	0.283***
	(0.01)	(0.02)	(0.11)	(0.05)	(0.09)	(0.05)
Individualism Square	-0.001***	-0.001***	-0.001*	-0.002***	-0.003***	-0.002***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Turning Point	66.52	65.76	67.51	63.66	68.20	68.22
County Group FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Year × State FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted-R^2	0.97	0.97				
1st Stage F			2.43	1.17	2.79	23.87
Observations	17631	17631	17631	17631	17631	17631
Panel B: Without County Group FE						
Individualism	0.188***	0.195***	0.249***	0.088	-0.071	0.602
	(0.02)	(0.02)	(0.04)	(0.82)	(0.60)	(0.40)
Individualism Square	-0.002***	-0.002***	-0.002***	-0.001	0.000	-0.005
	(0.00)	(0.00)	(0.00)	(0.01)	(0.00)	(0.00)
Controls	No	Yes	Yes	Yes	Yes	Yes
Turning Point	62.10	62.92	64.01	44.12	280.74	65.52
County Group FE	No	No	No	No	No	No
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted-R^2	0.93	0.93				
1st Stage F			2.23	0.18	0.17	0.37
Observations	17631	17631	17631	17631	17631	17631

Note: The dependent variable is log GDP per capita. The sample covers 1850–2010. Panels A and B report quadratic specifications with different fixed effects. Panel A reports specifications with county-group fixed effects, while Panel B reports specifications without county-group fixed effects but with state fixed effects. Columns (1) and (2) are OLS estimates, where column (2) adds population density as the control. Columns (3)–(6) are IV estimates using, respectively, the stock-based instrument, the transportation-based instrument, the interaction of transportation and KK, and the interaction of transportation and IHD7. The row labeled "Turning Point" reports the implied turning point from the estimated quadratic specification. The row labeled "1st Stage F" reports the first-stage F-statistic for the corresponding IV specification. Standard errors are clustered at the county-group and year levels. $^*p < 0.10$, $^{**}\ p < 0.05$, $^{***}p < 0.01$.

B.2.2 Innovation

Table A13 reports quadratic specifications for patenting. Compared with economic development, the evidence of nonlinearity is weaker and less consistent. While some specifications suggest nonlinear patterns, the estimates are sensitive to the choice of instrument. Overall, the results do not provide robust evidence of a stable nonlinear relationship between individualism and innovation.

Table A13: Individualism and Log Patent per capita within the US

	Log Patent per 1000 people (1850-2010)					
	OLS		IV: Stock	IV: Trans.	IV: Trans. × KK	IV: Trans. × IHD7
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: With County Group FE						
Individualism	0.126***	0.121***	-2.267**	0.238	0.735	0.360
	(0.03)	(0.03)	(1.05)	(0.19)	(0.57)	(0.37)
Individualism Square	-0.001***	-0.001***	0.016**	-0.002	-0.006	-0.003
	(0.00)	(0.00)	(0.01)	(0.00)	(0.00)	(0.00)
Turning Point	64.22	63.88	71.52	56.72	62.47	59.97
County Group FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Year × State FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted-R^2	0.83	0.83				
1st Stage F			2.88	0.49	0.73	8.24
Observations	12591	12591	12591	12591	12591	12591
Panel B: Without County Group FE						
Individualism	0.421***	0.423***	-0.180	-0.561	-1.443	2.676
	(0.07)	(0.07)	(0.36)	(2.50)	(3.46)	(2.77)
Individualism Square	-0.003***	-0.003***	0.001	0.003	0.009	-0.020
	(0.00)	(0.00)	(0.00)	(0.02)	(0.03)	(0.02)
Controls	No	Yes	Yes	Yes	Yes	Yes
TurningPoint	60.72	60.82	111.11	109.60	81.16	66.32
County Group FE	No	No	No	No	No	No
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted-R^2	0.41	0.41				
1st Stage F			2.27	0.62	0.42	0.26
Observations	12591	12591	12591	12591	12591	12591

Note: The dependent variable is log patent per 1000 people. The sample covers 1850–2010. Panels A and B report quadratic specifications with different fixed effects. Panel A reports specifications with county-group fixed effects, while Panel B reports specifications without county-group fixed effects but with state fixed effects. Columns (1) and (2) are OLS estimates, where column (2) adds population density as the control. Columns (3)–(6) are IV estimates using, respectively, the stock-based instrument, the transportation-based instrument, the interaction of transportation and KK, and the interaction of transportation and IHD7. The row labeled "Turning Point" reports the implied turning point from the estimated quadratic specification. The row labeled "1st Stage F" reports the first-stage F-statistic for the corresponding IV specification. Standard errors are clustered at the county-group and year levels. $^*p < 0.10$, $^{**}\ p < 0.05$, $^{***}p < 0.01$.

B.3 Additional Evidence on Time Variation

B.3.1 Economic Development

Figure A11 examines the time-varying influence of individualism on GDP per capita using fixed individualism categories. Combined with Figure 5, the results in Figure A11 show that the high individualism group is catching up in economic development and eventually becomes the group with the highest log GDP per capita, even though it exhibited the lowest level of log GDP per capita in the beginning.

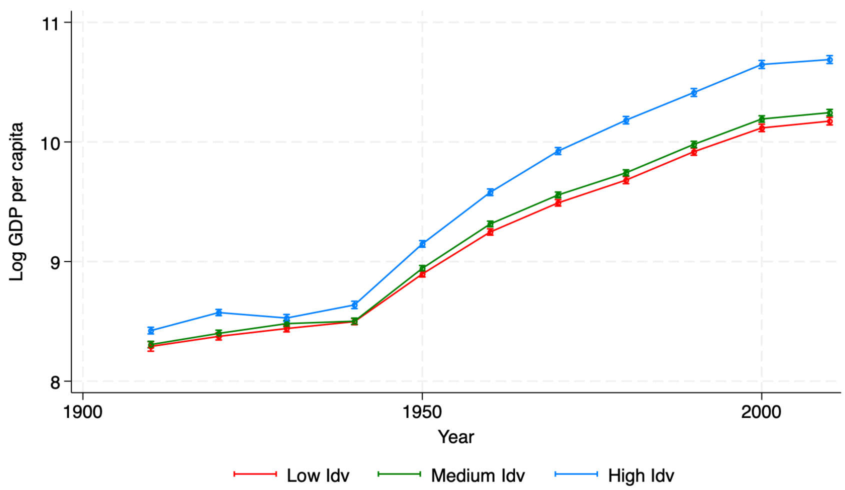


Figure A11: Individualism's Influence on Economic Development within the US over time (Category Fixed at 1900)

B.3.2 Innovation

Figure A12 presents the results for patenting activity. The low individualism group exhibited a higher increase in innovation before 1930. After 1930, the high individualism group caught up with a higher increasing rate but fell after 2000.

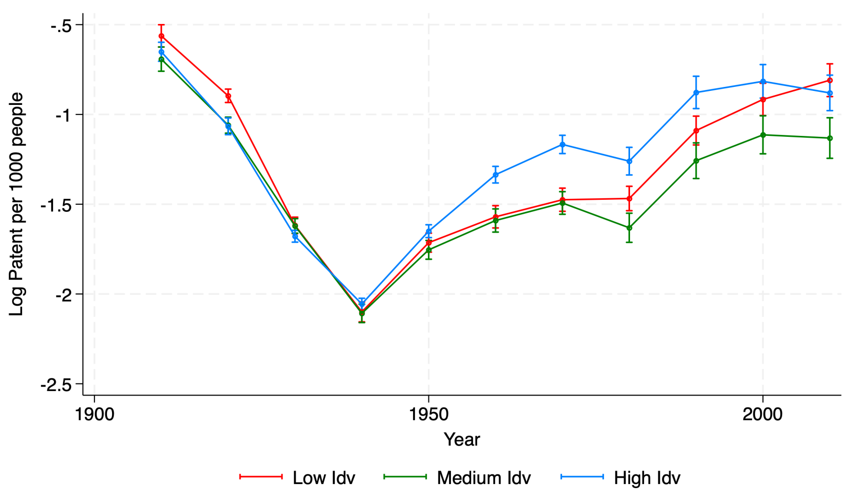


Figure A12: Individualism's Influence on Patents within the US over time (Category fixed at 1900)